## RSA and ECC Implementation

Objective

- Generate keys
- Encrypt and decrypt a message
- Understand modular arithmetic in practice

## PART 1: RSA Implementation

**Given:** $p = 13$, $q = 19$, $M = 9$, $e = 5$

- **Task 1: Key Generation**
  - Compute $n = p \times q$ and $\varphi(n) = (p-1)(q-1)$.
  - Find d such that $d \times e \equiv 1 \bmod \varphi(n)$.
- **Task 2: Encryption:** Compute ciphertext using: $C = M^e \bmod n$.
- **Task 3: Decryption:** Compute plaintext using: $M = C^d \bmod n$.

**Reflection Questions**

1. Why must e be relatively prime to φ(n)?
2. What are risks of small primes?

## PART 2: ECC Implementation Exercise

GIVEN:

Elliptic Curve: $y^2 = x^3 + 2x + 3 \bmod 17$
Base Point: $P = (5, 1)$
$d = 3$

- **Task 1: Key Generation:** Compute Public Key (Q).

**Reflection Questions**

1. Why is ECC secure with smaller keys?
2. How does modular arithmetic affect results?

## PART 3: AES Implementation Exercise

Show the complete solution of the first round:

| Round Number | Start of Round | After SubBytes | After ShiftRows | After MixColumns | Round Key Value |
|---|---|---|---|---|---|
| input | 32 88 31 e0<br>43 5a 31 37<br>f6 30 98 07<br>a8 8d a2 34 | | | | 2b 28 ab 09<br>7e ae f7 cf<br>15 d2 15 4f<br>16 a6 88 3c |
| 1 | 19 a0 9a e9<br>3d f4 c6 f8<br>e3 e2 8d 48<br>be 2b 2a 08 | d4 e0 b8 1e<br>27 bf b4 41<br>11 98 5d 52<br>ae f1 e5 30 | d4 e0 b8 1e<br>bf b4 41 27<br>5d 52 11 98<br>30 ae f1 e5 | 04 e0 48 28<br>66 cb f8 06<br>81 19 d3 26<br>e5 9a 7a 4c | a0 88 23 2a<br>fa 54 a3 6c<br>fe 2c 39 76<br>17 b1 39 05 |
| 2 | a4 68 6b 02<br>9c 9f 5b 6a<br>7f 35 ea 50<br>f2 2b 43 49 | 49 45 7f 77<br>de db 39 02<br>d2 96 87 53<br>89 f1 1a 3b | 49 45 7f 77<br>db 39 02 de<br>87 53 d2 96<br>3b 89 f1 1a | 58 1b db 1b<br>4d 4b e7 6b<br>ca 5a ca b0<br>f1 ac a8 e5 | f2 7a 59 73<br>c2 96 35 59<br>95 b9 80 f6<br>f2 43 7a 7f |
| 3 | aa 61 82 68<br>8f dd d2 32<br>5f e3 4a 46<br>03 ef d2 9a | ac ef 13 45<br>73 c1 b5 23<br>cf 11 d6 5a<br>7b df b5 b8 | ac ef 13 45<br>c1 b5 23 73<br>d6 5a cf 11<br>b8 7b df b5 | 75 20 53 bb<br>ec 0b c0 25<br>09 63 cf d0<br>93 33 7c dc | 3d 47 1e 6d<br>80 16 23 7a<br>47 fe 7e 88<br>7d 3e 44 3b |